

Statistics for Computer Science

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1 Introduction

This worksheet is intended to remind you of some basic data handling, probability and statistics that will be useful in your understanding of Computer Science topics and Artificial Intelligence specifically.

This workbook will not make you an expert in statistics, but will introduce some basic principles and give you an understanding of their use in Computer Science and hopefully give you the skills to delve deeper into the subject on your own, should that be of interest.

This material is intended to accompany a taught lecture and provides a number of questions with spaces for you to write your answers and check your understanding.

Why do we want to study statistics?

Statistics underpins many of the techniques we will use in machine learning and artificial intelligence and so, therefore, it makes sense to have an understanding of statistical concepts, even at a basic level. Furthermore the world into which we deploy our AI-enabled systems is uncertain and ever changing. Statistics gives us the tools to understand this world and hence design better systems and understand the ways in which they fail.

We might say that statistics is the language of data science.

The rest of this workbook is structured as follows. First we will take a look at *descriptive statistics* and *data visualization* that is techniques which help us understand the data we encounter. Next we will look at some core concepts in *probability*. Finally we will examine *data correlation* and *regression* which are practical links to machine learning.

2 Descriptive Statistics and Data Visualization

Consider the following: A survey was carried out to find out the ages of people visiting a news website. One hundred people were surveyed and the ages recorded are presented in a table below with results placed into ascending order for convenience.

Now with a table of this size it is possible that we could inspect each element individually and painstakingly uncover the secrets held within it but if this was a survey of 60,000 people then this would be impractical. But even with the modest list presented here I hope you agree that it's hard to see what is going on from the raw data presented in this way

19	19	19	19	20	20	21	21	21	23
24	24	25	25	26	26	28	28	29	31
31	32	32	33	34	35	35	36	37	38
38	38	39	39	40	41	41	41	42	42
43	43	44	45	45	46	46	47	50	52
53	53	54	55	56	56	56	57	57	59
59	60	61	61	61	61	64	64	64	66
67	67	68	68	69	69	70	70	70	71
71	72	74	75	75	75	76	77	77	77
77	78	78	78	79	79	79	79	80	80

A first step in understanding the table may be to group data. The question we want to ask of the data informs how that grouping might happen.

“Is our website equally popular across all age groups.”.

In this case then we might choose a number of age groups as:

$$(18 - 29), (30 - 42), (43 - 54), (55 - 67), (68 - 80)$$

and then summarise our data into a smaller set of values as shown below.

Age Group	Frequency
18-29	19
30-42	21
43-54	13
55-67	19
68-80	28

We might then make use of a visualization to show these results. In this case a histogram seems like a sensible choice and a plot are shown in Figure 1. Looking at the data in this way shows us that we attract many people in the oldest group but we are not successful in reaching people in the 43-54 age range.

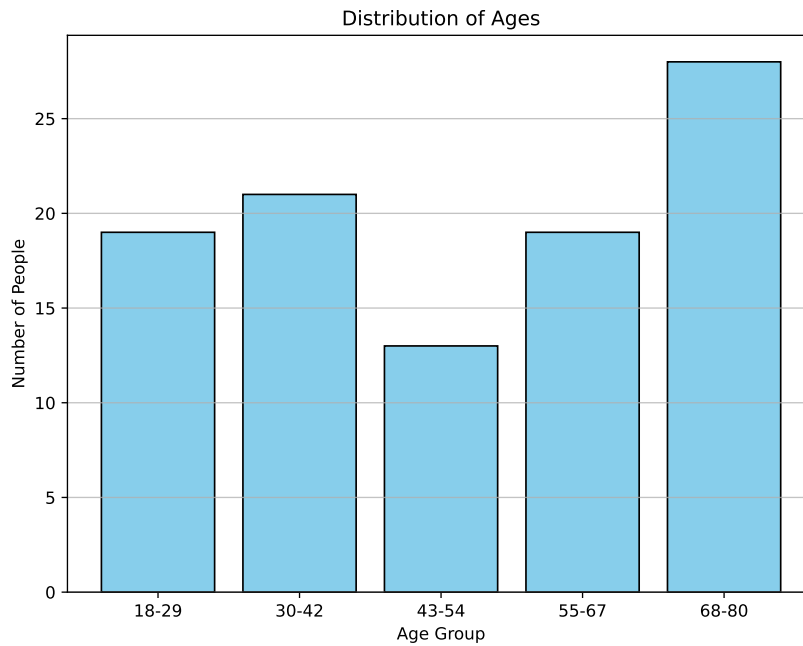


Figure 1: Histogram showing the results of our age survey

2.1 Central Tenancy

A common requirement in handling data is to find out what the average “thing” is. I’m being deliberately vague here, but it might be the average age of a patient, the aver temperature at which failure occurs for a jet engine etc.

Consider a machine which makes ball bearings. We know they are all very similar but no two ball bearings are identical. If we consider their weight we could describe all the ball bearings as have

$$\text{weight} = \bar{w} \pm \epsilon$$

where $\bar{w}$ is the weight we designed the machine to create and ϵ is some measure of variance on that value.

Of course if we were involved in the design of the machine we might know precisely what $\bar{w}$ was but if the process is natural finding this value hidden by noise is more tricky.

The most common measures one would use to find this value are the **mean**, **mode** and **median**. Each of these are considered to be measure of central tendency.

Let’s consider the following set of data which reports the number of snails found in 31 gardens one a one hour time period. Rather than providing the raw data as before we now present the data as a frequency table, Table 1.

Number of Snails	0	1	2	3	4	5	24
Frequency	3	5	11	8	2	1	1

Table 1: Number of Snails found in a garden

To better visualise this data we will again create a histogram, Figure 2. This data looks quite different from the first histogram we created. Whilst the first had a roughly flat *distribution* across the groups this second plot shows a distinct peak meaning it's most common to see 2 snails and the likelihood of seeing more or less snails reduces as we move away from this point. We also see our first example of an outlier. An outlier is a sample (or data point) which doesn't look like the rest (24 snails in one garden).

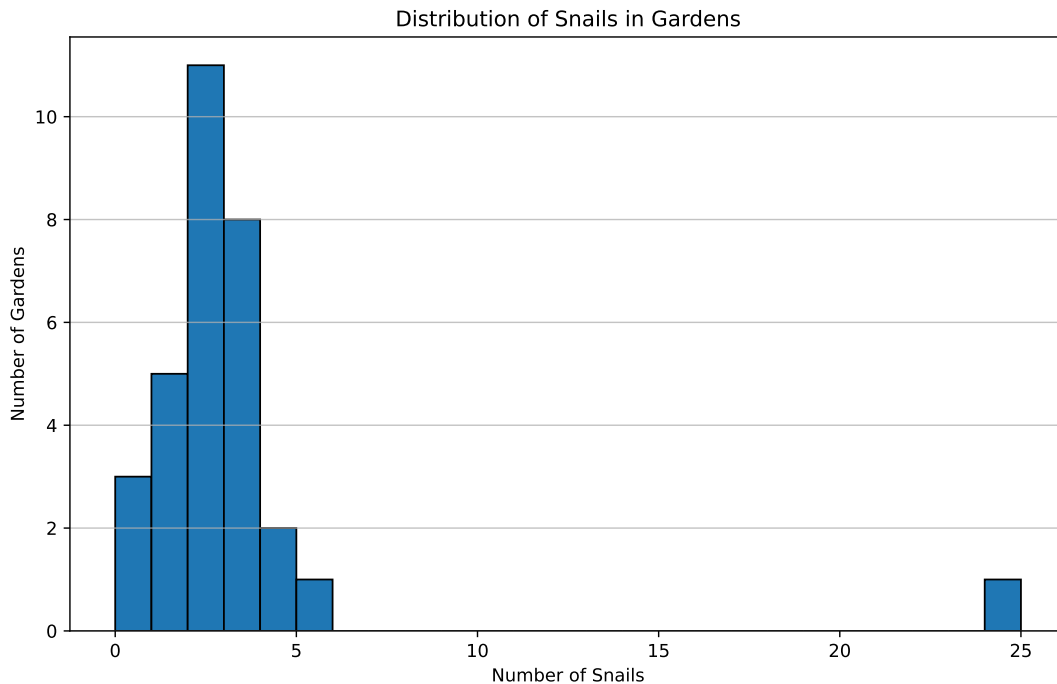


Figure 2: Number of snails found in gardens

Question 1.

What might be the cause of the outlier?

So let's start with our question:

“How many snails, on average, will you find in a garden”

The most common ways of answering this question is with one of three measures of central tendency, Mode, Median and Mean. All of these measures are designed to tell us something about the “centre” of the data.

Mode is simply the “most common” number in the data. For our data we simply look for the number with the highest frequency which is 2, i.e. 11 gardens reported 2 snails. This means that if we put all 31 numbers into a bag and picked one at random the number we are most likely to see is 2.

Median is the middle number when all the numbers are placed in numeric order. For our data we would then write the data out as follows:

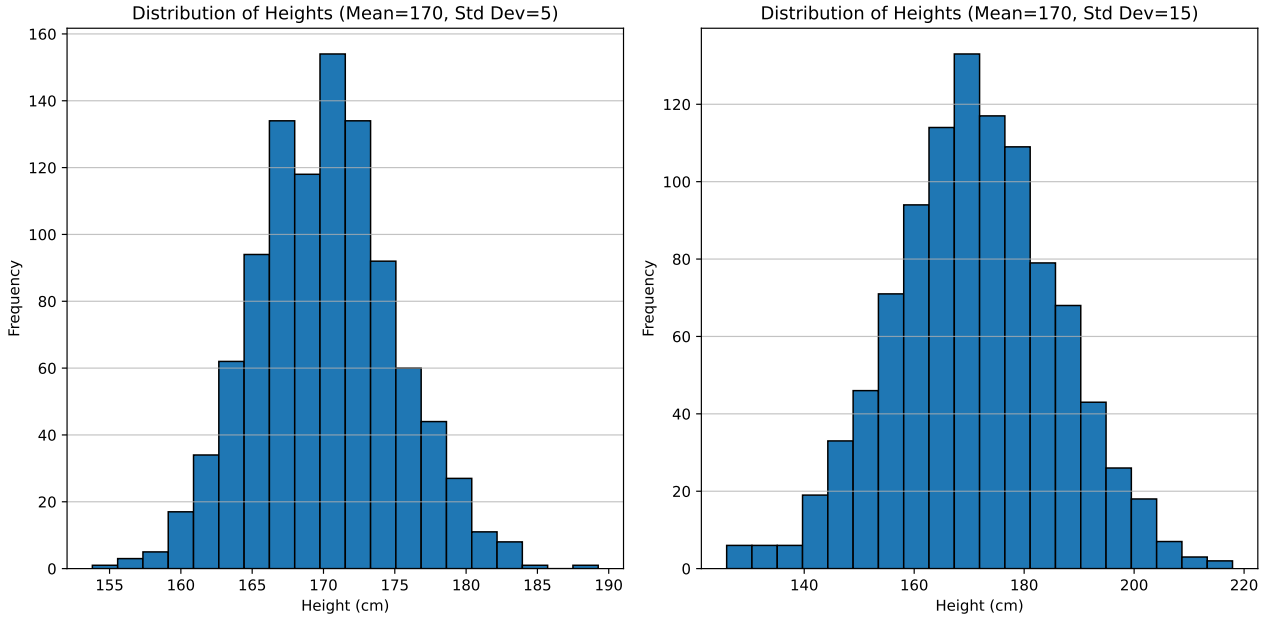


Figure 3: Two histograms showing the heights of individuals with the same mean but different variance

An extension to this idea is to plot outliers, as circles. There are a number of ways in which we might calculate outliers but I leave this to the interested reader to research for themselves.

2.4 Standard Deviation

A second method for describing the *dispersion* of a dataset is called the standard deviation and it is calculated as follows:

$$s = \sqrt{\left(\frac{\sum_{i=1}^n (x_i^2)}{n} - \bar{x}^2\right)} \quad (3)$$

I know that looks really horrible but let's break it down. x_i is each of the values we have recorded in our experiment (in this heights). So we will square each height in the list and then add them together (that's what the $\sum$ symbol means). Next we will divide that number by n which is the number of heights we recorded. $\bar{x}$ is just maths notation for saying the **mean** of x , which we showed how to calculate above in Equation 2.

So this equation then says the standard deviation (s) is the square root ($\sqrt{()}$) of the mean of squares ($\frac{\sum_{i=1}^n (x_i^2)}{n}$) minus the square of the mean ($\bar{x}^2$).

Let's give it a go on a small set of data

x	x^2
2	4
3	9
5	25
6	36
8	64
$\bar{x} = 4.8$	$\sum(x^2) = 138$

so

$$s^2 = \frac{138}{5} - (4.8)^2 = 4.56$$

$$s = \sqrt{4.56} = 2.14$$

So what does this number actually tell us? Well without going into too much detail it says the majority of the data will lie within 2 standard deviations of the mean. So looking once again at our histograms in Figure 3 we

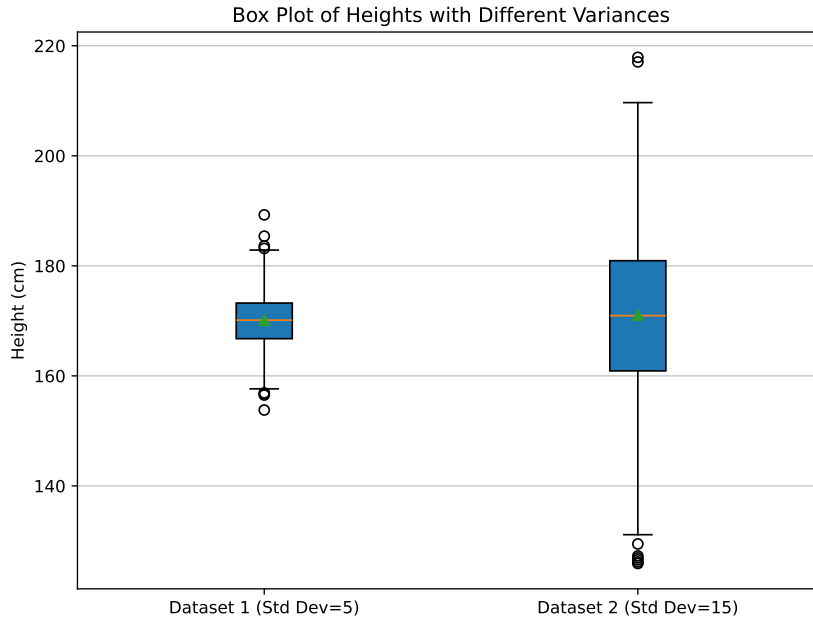


Figure 4: A box plot of the two heights data sets

see that most of the data should lie within the range 170 ± 10 for the left hand graph and within 170 ± 30 for the right hand graph. **Is this what you see?**

Question 3.

We sampled the output from two machines which create biscuits. For each biscuit we measured it's weight. Calculate the mean and standard deviation of biscuit weights for each machine and then comment on the performance of each machine^a.

Machine A

196	198	198	199	200	200	201	201	202	205
-----	-----	-----	-----	-----	-----	-----	-----	-----	-----

Machine B

192	194	195	198	200	201	203	204	206	207
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^aQuestion taken from A concise course in Advanced Level Statistics with worked examples. J. Crawshaw and J. Chambers.

3 Probabilities

The nature of a chance event is that we can not say what will happen, but for some systems, and with appropriate assumptions, we can predict accurately the *proportion* of times an event would occur.

Consider a normal six sided dice. If I throw the dice 600 times how many sixes am I likely to get?

For our dice for example we can reasonably say that all of the numbers are equally likely (assuming a fair dice)

and that therefore we will get a 6 (or any other number) $\frac{1}{6}$ of the time we roll our dice.

Question 4.

Take a coin and flip it 10 times. Fill in the table below with pHeads and pTails being the proportion of heads and tail you have flipped so far.

Flip No.	Outcome	pHeads	pTails
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			

I ran the same experiment but repeating the experiment 30 times. My results are shown as a graph in Figure 5. The first few flips were as follows: $\{T, H, T, T, H, H, \dots\}$

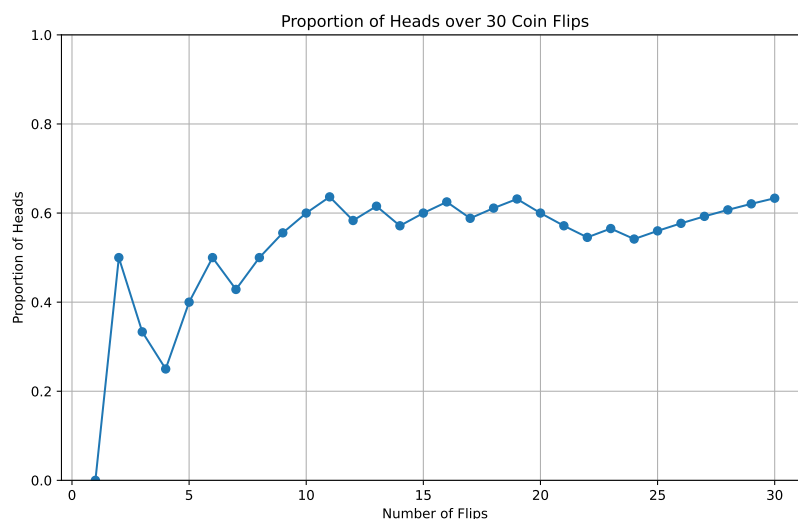


Figure 5: Coin fLip experiment showing the proportion of Heads

Notice that at the beginning the proportion varies greatly but as the experiment continues the variation decreases. Notice also that we also still seem to be quite a way off of 0.5! If I run the experiment again, Figure 6 this time for 3000 flips we can see the value setting to the number we expect. However we see in this experiment that the initial estimate is very high.

When we write a probability then we are saying the proportion of times we would expect and event to occur if the event were repeated infinitely often. For our example the event E is:

$$E = \text{coin lands on heads}$$

and the probability of the event is:

$$P(E) = 0.5$$

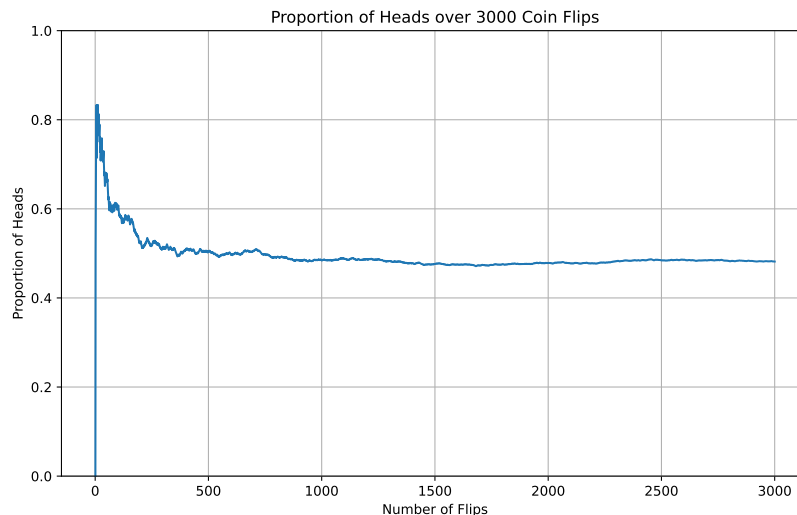


Figure 6: 3000 Coin flips

Question 5.

1. Can you think of a way of describing the following probabilities using plain English statements.

- $P(E) = 0$
- $P(E) = 1$

2. Can a probability be negative?

A quick note. **Events** and **Outcomes** are not the same thing. Consider a dice roll the outcomes are six different numbers we can get. The event however is the subset of outcomes we are interested in. So each of the following could be an event:

1. Rolling a six : $P(E) = \frac{1}{6}$
2. Rolling an even number : $P(E) = \frac{1}{2}$
3. Rolling a number greater than 4 : $P(E) = \frac{1}{3}$

Aside: Although we have started this discussion of probabilities using dice and coins it is worth noting that probabilities are used in situations where calculating probabilities in advance of the experiment is not quite so easy.

Consider the following examples.

1. **Weather:** Will it snow this Christmas?
2. **World Events:** Will the U.S.A. have another Civil War?

Sometimes probabilities can be based on the frequency of easily observed events (or easily modelled processes) but sometimes probabilities are a way of communicating our own beliefs. For both cases probabilities are often stated without making it clear what assumptions are made, this can be problematic. For example is the frequency of previous snow a good model for future snow?

3.1 Multiple events

Often we are interested in not just one event but multiple events happening either at the same time or in sequence.

For a moment think about how would you might approach calculating the following probabilities.

1. Getting two heads when flipping two coins
2. Rolling three sixes on three dice
3. The probability of getting at least one six when 3 dice are rolled
4. If we select 3 cards at random from a standard deck of cards at least one of them will be a picture card?

Two Coins

One way to approach this problem is to list all the possible outcomes and then work out the proportion which match our desired event.

- Head, Head
- Head, Tail
- Tail, Head
- Tail, Tail

Notice that it is importance to be systematic in writing out the possible outcomes. Here it is clear that the probability of two heads is $P(E) = 0.25$ but if I had written $\{2Heads, 2Tails, 1Headand1Tail\}$ as my list of outcomes I may have missed that there are four different combinations possible.

3 Dice

Whilst the approach shown above works for a small number of outcomes it completely runs out of steam when the number of outcomes increase. For our 3 dice problem there are 216 ($6 \times 6 \times 6$) different outcomes, I really don't want to write all of these out.

Fortunately I can use a **multiplication** rule for this problem because I know that rolling one dice does not affect the next roll, i.e. the events are independent.

This means that we can calculate the total probability of each event (getting a six) and then multiplying these together so that

$$P(3 \text{ sixes}) = \frac{1}{6} \times \frac{1}{6} \times \frac{1}{6} = \frac{1}{216}$$

So generally if we can be sure that two events (E and F) are independent we can simply multiply the event probabilities i.e. $P(E \text{ and } F) = P(E) \times P(F)$.

Question 6.

Calculate the probability of obtaining a total of 3 in each of two successive rolls of 2 dice.

Step 1 : Probability of rolling 3 on two dice

Step 2 : Calculate the probability of doing this twice in a row.

At least 1 six

Probabilities always add up to one so we can use this to answer this question by saying we want 1 minus the probability of getting at NO sixes.

$$P(\text{no sixes}) = \frac{5}{6} \times \frac{5}{6} \times \frac{5}{6} = 0.579$$

Now we know that all other options will have at least one 6 so:

$$P(\text{at least one six}) = 1 - P(\text{no sixes}) = 1 - 0.579 = 0.421$$

If we consider ALL possible events then the probabilities of these events MUST add to 1. To demonstrate this let's introduce a graphical device called a tree diagram. Figure 7 shows all possible outcomes for an experiment in which we first flip a coin and then roll a dice to see if we get a six.

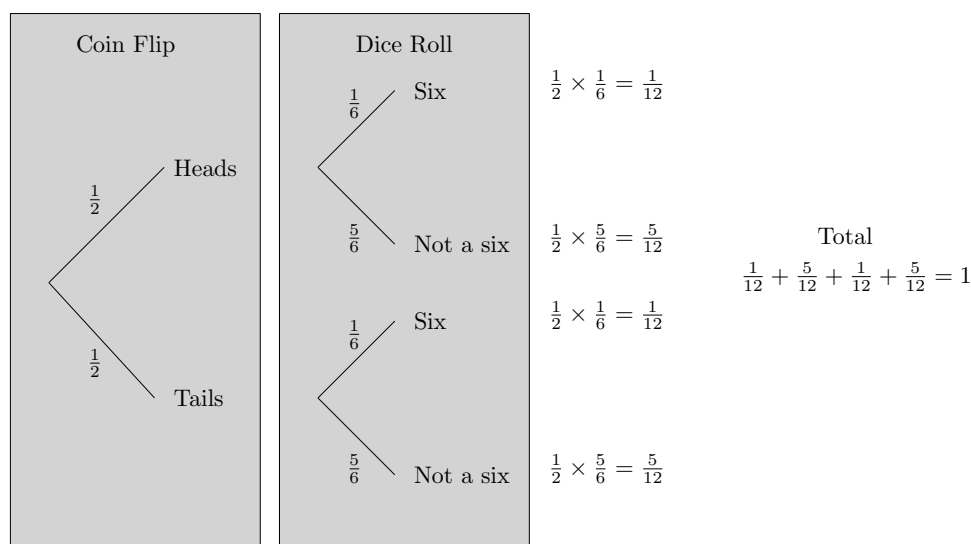


Figure 7: Example of a tree diagram to examine the probabilities of two events

For each experiment we walk along a branch of the tree multiplying the probabilities as we go. If we then add up the final outcomes we get a sum of 1.

Picture Cards

So far every experiment has made use of the fact that the outcomes are independent, but what if they are not. Well let's start by thinking about the Picture Card problem and what might happen.

If we select 3 cards at random from a standard deck of cards at least one of them will be a picture card?

Why isn't this problem just like the others? Well once we take a card out of the deck then the deck has changed. The first action we took impacts the probability of the second. So that means that

The first card we draw will be a picture card with probability $\frac{3}{13}$

But this isn't our only chance to draw a picture card so now we need to work out the following ...

Question 7.

What is the probability of drawing a picture card with the second card?

Regardless of what we drew the first time we only have 51 cards in the deck. But also if we picked a picture card the first time then the number of picture cards will go down.

Now this is another example of where we might use our trick of looking for the opposite of what we want as it reduces the number of options. So Let's change our question a little :

If we select 3 cards at random from a standard deck of cards *NONE* of them will be a picture card?
 So now the probability of not drawing a picture card each time is as follows:

$$P(\text{no pic}) = \frac{40}{52} \times \frac{39}{51} \times \frac{38}{50} = 0.447$$

Now we can answer our original question by assuming all other events will include a picture card.

$$P(\text{pic}) = 1 - P(\text{no pic}) = 1 - 0.447 = 0.553$$

3.2 Bayes Theorem

This last example introduces us to the idea of asking a probabilistic question when we know the outcome of a previous probabilistic event. Why is this important? Well consider a medical scenario in which we can observe symptoms and we want to assess the probability of a disease. In Natural language we might want to write something like:

The probability of someone having a cold given that they have a runny nose is 0.4 whilst the probability of them have cold if they don't have a runny nose is 0.2.

Mathematically you would write the statement “The probability of A given the event B happened” as:

$$P(A|B)$$

This is known as a conditional probability since the probability of A is a condition of B. For our case we might write:

$$P(\text{cold}|\text{runny nose}) = 0.4$$

and

$$P(\text{cold}|\neg\text{runny nose}) = 0.2$$

notice that these don't add up to one because that's not all the possible outcomes and combinations.

One question of course is where did those numbers come from? In this case it might be experimental (or observed data) but more generally it may not be possible to have numbers directly for these conditional probabilities.

Consider the following statements:

- *The probability of a particular form of cancer being present is 0.01*
- *The probability of a positive test when cancer present is 0.9.*
- *The probability of a positive test when cancer not present is 0.05.*

But our question is “if you test have a positive cancer test how likely is it that you have cancer?”. We don't have a value for this.

Writing this formally we want to know:

$$P(\text{cancer}|\text{positive})$$

Fortunately there is a way of calculating this value and it uses something called “Bayes Rule” as shown in Equation 4:

$$P(A|B) = \frac{P(A)P(B|A)}{P(B)} \tag{4}$$

Let's now plug in our values and see what we get

$$P(\text{cancer}|\text{positive}) = \frac{P(\text{cancer})P(\text{positive}|\text{cancer})}{P(\text{positive})}$$

Looking at our data in above we see that:

- $P(\text{cancer}) = 0.01$
- $P(\text{positive}|\text{cancer}) = 0.9$

We still need to know $P(\text{positive})$, the probability of receiving a positive test result. Let's consider 10,000 patients selected at random.

Question 8.

Given the probability data above, answer the following questions:

- a = Out of 10,000 people how many will have cancer
- b = How many with cancer will get a positive test
- c = How many without cancer will get a positive test

What is the value of $P(\text{positive}) = \frac{(b+c)}{10,000}$.

Using this calculated value, what is $P(\text{cancer}|\text{positive})$

How would you interpret this value?

4 Data correlation and regression

One of the problems we are often looking at in AI is how variables are related. We want to know if a given factor has an affect on an outcome, or to put it in simpler terms is a thing we can control will change something we are interested in and if so how.

So we might ask questions like does the speed we drive at impact the severity of injuries in car accidents, or does the amount of exercise we undertake impact bone health in later life.

Questions like this require us to consider if variables are **correlated**.

Let's start by looking at some data as shown in Figure 8. You might imagine here that x is age and y is salary for example or amount of exercise and additional years of healthy life.

Visually we can see that as x increases so does y. This is known as a positive correlation.

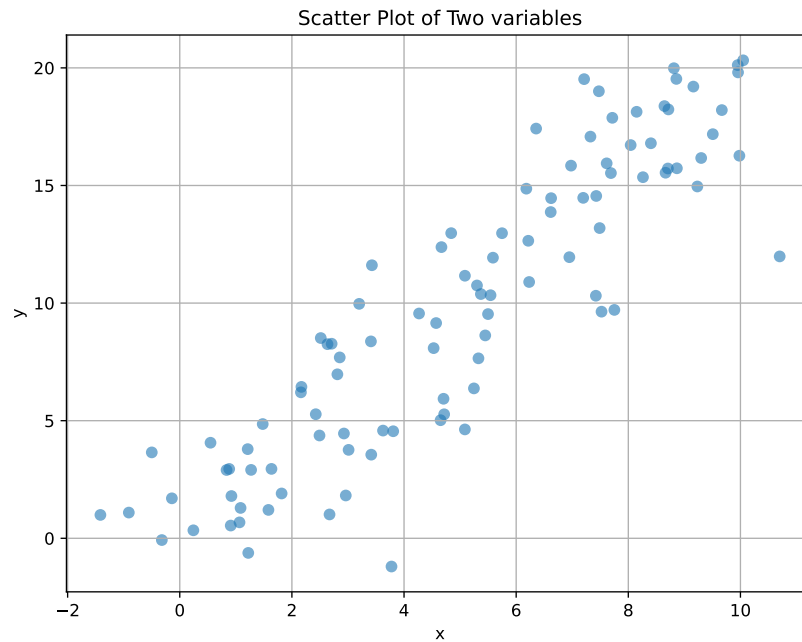


Figure 8: Example of a positive correlation between two variables.

Question 9.

It has been shown that butter consumption is positively correlated with ticket prices at North American movie theatres^a.

Can you explain this correlation?

^ahttps://www.tylervigen.com/spurious/correlation/1604_butter-consumption_correlates-with_ticket-prices-at-north-american-movie-theaters

When we look at a graph such as that shown in Figure 8 we are often interested in the function which describes one variable in terms of the other. In this case, for example it might seem reasonable to assume that there is an underlying linear process to which noise has been added. Indeed this is how I generated the data!

If there were no noise then a straight line plot is simply described as:

$$y = mx + c$$

where m is the gradient of the line and c is the point where the line crosses the y axis. So for the previous data this might look like the line shown in Figure 9.

Question 10.

Using a pen and ruler draw a line of best fit on the two scatter plots shown in Figure 10

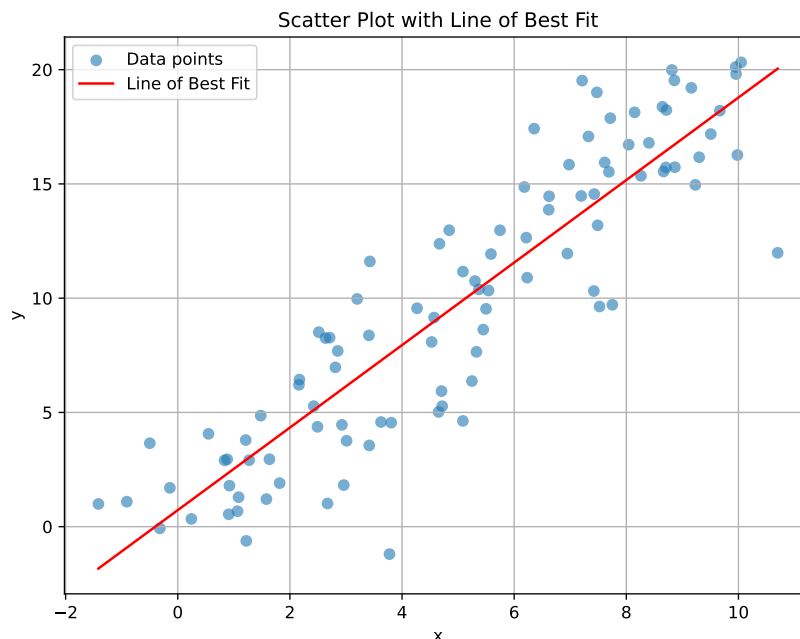


Figure 9: A scatter plot with a line of best fit

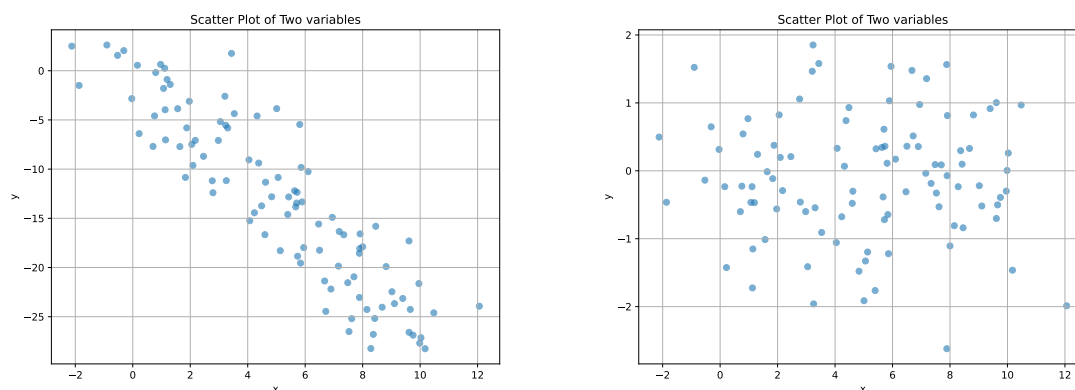


Figure 10: Correlation Plots

4.1 Least Squares Regression

To round off this course we are going to look at a method of calculating the best fit line using something called least squares regression.

Let's start by looking at a small set of data as shown in Table 2:

Time, x (min)	5	7	12	16	20
Mass, y (g)	4	12	18	21	24

Table 2: Weight of a chemical over reaction time

We will start with a scatter plot of the data as shown in Figure 11. Here we have drawn a line $y = 1.6x$. We don't know if this is the "best fit" but it's not a bad guess. What we can now do is measure the distance of each point (vertically) from the line we have guessed (e_i). We think of these measurements as the *error* between our guess (line) and the observed data. The total error is then:

$$\epsilon = \sum_i^n e_i$$

where n is the number of data points (n).

It seems quite logical therefore to say that a good guess for the line would have a very small value of ϵ whilst a poor guess would have a large value.

Our job then is to find the line for which ϵ is as small as possible.

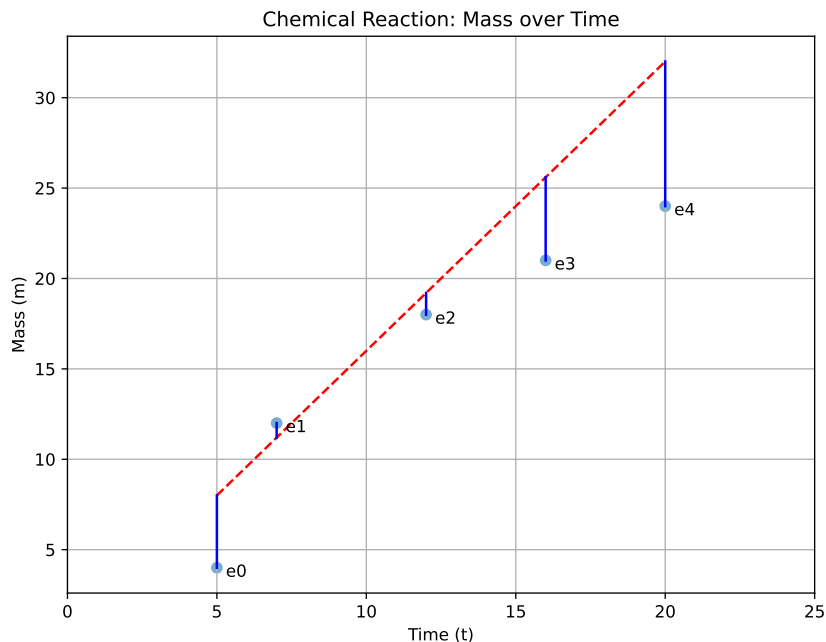


Figure 11: Mass of chemical over time

So how might we calculate this line of best fit. The process is to first think of a function that is a reasonable guess. For us this is a straight line, and the formula for this is commonly written as :

$$y = mx + c$$

So now we have to work out the values of m and c that give us the smallest error possible. One way we could do this is guess values and continue guessing randomly for 100 tries and choose the answer with the smallest value, but this seems a little inefficient.

In your AI course you will come across ways to do this for complex functions but for now we will look at a method using maths.

We know that gradient of a line b is the change in y divided by the change in x . What is maybe not so clear is that we can also calculate the gradient as

$$m = \frac{s_{xy}}{s_{xx}}$$

where s_{xx} is the variance of the values of x and s_{xy} is something called covariance.

Variance is a like the standard deviation we calculated in Equation 3.

$$s_{xx} = \frac{\sum_{i=1}^n x_i^2}{n} - \bar{x}^2 \quad (5)$$

Note then that $s_{xx} = s^s$ and is similarly a measure of the “spread” of the variable x .

Covariance is still a measure of *joint* variability and is calculated as follows:

$$s_{xy} = \frac{\sum_{i=1}^n x_i y_i}{n} - \bar{x} \bar{y} \quad (6)$$

So for our problem we have 5 data points let’s make a small table to help up with our calculation of b .

Question 11.

Complete Table 3 and put the sum of each column in the final row.

x	y	x^2	y^2	xy
5	4	25	16	20
7	12			
12	18			
16	21			
20	24			
$\Sigma = 60$	$\Sigma = 79$			

Table 3: Calculation table for variance and co-variance

Now we can calculate s_{xx} . We know that the average of x is $\bar{x} = \frac{60}{5}$ so,

$$s_{xx} = \frac{874}{5} - \left(\frac{60}{5}\right)^2 = 30.8$$

Similarly:

$$s_{xy} = \frac{1136}{5} - \frac{60}{5} \frac{79}{5} = 37.6$$

so the gradient of the correct line is

$$b = \frac{37.6}{30.8} = 1.22$$

Great we have a gradient, how do we now find c ? Well the trick is to know that the point $(\bar{x}, \bar{y})$ lies on the line. So we can rearrange our equation for a straight line to say:

$$c = \bar{y} - m\bar{x} = \frac{79}{5} - 1.22 \frac{60}{5} = 1.16 \quad (7)$$

Question 12.

This felt like a lot of work so why might we want to work out the equation of a function that fits our data?

Of course just like other problems we have seen in this course we can get the computer to help us ...

1. Go to www.wolframalpha.com
2. Click on the examples button
3. Find the examples for Statistics
4. Find the Regression Analysis Examples
5. Click on the example starting "linear fit"
6. Click on the "=" button and explore the results
7. Try entering the data from our problem.
8. Do you get the same answer?

5 Suggested Books

A concise course in Advanced Level Statistics (with worked examples) J. Crawshaw and J. Chambers: (Available from the University Library).

A Modern Introduction to Probability and Statistics: Understanding Why and How: Dekking et al. : <https://link.springer.com/book/10.1007/1-84628-168-7>

Statistics for Research Students : Fein et al. <https://usq.pressbooks.pub/statisticsforresearchstudents/>

Understanding Statistics and Experimental Design: How not to lie with statistics: Herzog et al. <https://link.springer.com/book/10.1007/978-3-030-03499-3>